

Introduction to Lie Group Theory

Tejaswi
Supervisor : Prof. Anupam Singh

Summer 2026

Contents

1	Matrix Groups	1
1.1	Groups of Matrices	1
1.2	Matrix Groups	2
1.3	Complex Matrices as Real Matrices	5
1.4	Matrix Groups of Normed Vector Spaces	5
1.5	Matrix Exponential and Logarithm	6
1.6	Differential Equations in Matrices	7
1.7	One Parameter Subgroups in Matrix Groups	8
2	Tangent Spaces and Lie Algebras	9
2.1	Lie Algebras	9
2.2	Curves and Tangent Spaces	9
2.3	Lie Algebras of Matrix Groups	9
2.4	$SO(3)$ and $SU(2)$	9
2.5	Complexification of Real Lie Algebra	9
3	Lie Groups	10
3.1	Smooth Manifolds	10
3.2	Tangent Spaces and Derivatives	10
3.3	Lie Groups	10
3.4	Matrix Groups are Lie Groups	10
3.5	Not All Lie Groups are Matrix Groups	10

Chapter 1

Matrix Groups

1.1 Groups of Matrices

Let $M_{m,n}(\mathbb{K})$ be the set of $m \times n$ matrices whose entries are in $\mathbb{K}$, a commutative field. We will also use the special notations $M_n(\mathbb{K}) = M_{n,n}(\mathbb{K})$, and $\mathbb{K}^n = M_{n,1}(\mathbb{K})$.

$M_{m,n}(\mathbb{K})$ is a vector space over $\mathbb{K}$ with dimension mn . As well as being a vector space over $\mathbb{K}$ with dimension n^2 , $M_n(\mathbb{K})$ is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensional algebra over $\mathbb{K}$.

We will use the notation

$$GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible $n \times n$ matrices, known as the $n \times n$ *general linear group*, and

$$SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq GL_n(\mathbb{K})$$

for the set of $n \times n$ unimodular matrices, known as the $n \times n$ *special linear group*.

Theorem 1.1. *The sets $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are groups under matrix multiplication. Furthermore, $SL_n(\mathbb{K})$ is a subgroup of $GL_n(\mathbb{K})$.*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continuous.

Proposition 1.1. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then

- (i) $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is an open subset;
- (ii) $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ is a closed subset.

Proof. $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$, which is open since $\{0\}$ is closed. Similarly, $SL_n(\mathbb{K}) = \det^{-1}\{1\}$, which is closed since the singleton $\{1\}$ is closed in $\mathbb{K}$. ■

Definition 1.1. Let G be a topological space, and view $G \times G$ as the product space. Suppose that G is also a group with multiplication map $\text{mult} : G \times G \rightarrow G$, and inverse map $\text{inv} : G \rightarrow G$. G is a *topological group* if mult and inv are continuous.

Theorem 1.2. For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are topological groups.

1.2 Matrix Groups

Definition 1.2. A subgroup G of $GL_n(\mathbb{K})$ is called a *matrix group* if it is a closed subset of $GL_n(\mathbb{K})$.

Remark. We may consider $GL_n(\mathbb{K})$ as a subgroup of $GL_{n+1}(\mathbb{K})$ by identifying the $n \times n$ matrix $A = [a_{ij}]$ with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that $GL_n(\mathbb{K})$ is closed in $GL_{n+1}(\mathbb{K})$. With the aid of this embedding, any matrix subgroup of $GL_n(\mathbb{K})$ can also be viewed as a matrix subgroup of $GL_{n+1}(\mathbb{K})$.

Definition 1.3. Let G, H be two matrix groups. A groups homomorphism $\varphi : G \rightarrow H$ is a *continuous homomorphism of matrix groups* if it is continuous and its image $\varphi(G)$ is a closed subgroup of H .

Examples. Some important examples of real and complex matrix groups:

1. Groups of Upper Triangular Matrices

For $n \geq 1$, an $n \times n$ matrix $A = [a_{ij}]$ is said to be *upper triangular* if $a_{ij} = 0$ if $i < j$. An upper triangular matrix is said to be *unipotent* if all of its diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of $GL_n(\mathbb{K})$ is

$$UT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is upper triangular}\},$$

while the *unipotent subgroup* of $GL_n(\mathbb{K})$ is

$$SUT_n(\mathbb{K}) = \{A \in GL_n(\mathbb{K}) : A \text{ is unipotent}\}.$$

It is easy to verify that $UT_n(\mathbb{K})$ and $SUT_n(\mathbb{K})$ are closed subgroups of $GL_n(\mathbb{K})$, and that $SUT_n(\mathbb{K})$ is a subgroup of $UT_n(\mathbb{K})$.

2. Affine Groups

The *n-dimensional affine group* over $\mathbb{K}$ is

$$Aff_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in GL_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

which is clearly a closed subgroup of $GL_{n+1}(\mathbb{K})$. If we identify $x \in \mathbb{K}^n$ with $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$, then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

we obtain an action of $Aff_n(\mathbb{K})$ on $\mathbb{K}^n$.

The group $Aff_n(\mathbb{K})$ can be expressed as the semi-direct product of $Trans_n(\mathbb{K})$ and $GL_n(\mathbb{K})$, where

$$Trans_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

is a closed normal subgroup of $Aff_n(\mathbb{K})$. That is,

$$Aff_n(\mathbb{K}) = GL_n(\mathbb{K}) \ltimes Trans_n(\mathbb{K}).$$

3. Orthogonal Groups

The *n × n real orthogonal group* is the subset

$$O(n) = \{A \in GL_n(\mathbb{R}) : A^T A = I_n\},$$

where A^T is the transpose of $A = [a_{ij}]$ whose entries are given by $(A^T)_{ij} = a_{ji}$. $O(n)$ is a closed subgroup of $GL_n(\mathbb{R})$. For all $A \in O(n)$, since $\det A = \det A^T$, $\det A = \pm 1$. The subgroup

$$SO(n) = \{A \in O(n) : \det A = 1\}$$

is known as the *special orthogonal group*. One of the main reasons for the study of orthogonal groups is due to their relation with isometries. If $A \in GL_n(\mathbb{R})$, then it can be proved that A is a linear isometry iff A is orthogonal. The elements of $SO(n)$ are called *direct isometries* or *rotations*.

4. Unitary Groups

The $n \times n$ *unitary group* is the subset

$$U(n) = \{A \in GL_n(\mathbb{C}) : A^*A = I_n\},$$

where A^* is the *hermitian conjugate* of A , that is $A^*)_{ij} = a_{ji}$. $U(n)$ is clearly a closed subgroup of $GL_n(\mathbb{C})$. The *special unitary group* is the subset

$$SU(n) = \{A \in U(n) : \det A = 1\},$$

which is also a normal subgroup of $U(n)$.

5. Symplectic Groups

A matrix S is called *skew symmetric* if $S^T = -S$. The standard example of this is built up using the 2×2 block

$$J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

When $m \geq 1$ we have the non degenerative skew symmetric matrix

$$J_{2m} = \begin{bmatrix} J & O_2 & \cdots & O_2 \\ O_2 & J & \cdots & O_2 \\ \vdots & \vdots & \ddots & \vdots \\ O_2 & O_2 & \cdots & J \end{bmatrix}.$$

The matrix group

$$\text{Symp}_{2m}(\mathbb{R}) = \{A \in GL_{2m}(\mathbb{R}) : A^T J_{2m} A = J_{2m}\}$$

is called the $2m \times 2m$ *real symplectic group*. It can be proved that if $A \in \text{Symp}_{2m}(\mathbb{R})$, then $\det A = 1$.

1.3 Complex Matrices as Real Matrices

Similar to how complex numbers can be viewed as a 2-dimensional real vector space with basis $(1, i)$, every $n \times n$ complex matrix can also be viewed as a $2n \times 2n$ real matrix using the following constructions or the ones similar to it.

We identify each complex number $z = x + iy$ with a 2×2 matrix by defining a function

$$\rho : \mathbb{C} \longrightarrow M_2(\mathbb{R}); \quad \rho(x + iy) = \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

This can also be expressed as

$$\rho(x + iy) = xI_2 - yJ.$$

As ρ is an injective ring homomorphism, we can view $\mathbb{C}$ as a subring of $M_2(\mathbb{R})$. Notice that $\rho(\bar{z}) = \rho(z)^T$.

For a $n \times n$ complex matrix Z , we define a function $\rho_n : M_n(\mathbb{C}) \longrightarrow M_{2n}(\mathbb{R})$;

$$\rho_n(Z) = \begin{bmatrix} \rho(z_{11}) & \rho(z_{12}) & \cdots & \rho(z_{1n}) \\ \rho(z_{21}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \rho(z_{n1}) & \cdots & \cdots & \rho(z_{nn}) \end{bmatrix}$$

Then ρ_n is an continuous injective ring homomorphism. Generally for $n \times n$ real matrices X and Y ,

$$\rho_n(X + iY) = \rho_n(X) - J_{2n}\rho_n(Y)J_{2n}.$$

1.4 Matrix Groups of Normed Vector Spaces

It is often desirable to generalize the notion of matrix spaces to arbitrary finite dimensional real or complex vector spaces. Here we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Theorem 1.3. *Let (U, μ) and (V, ν) be two finite dimensional normed vector spaces of the same dimension over $\mathbb{K}$. Then any linear isomorphism $\varphi : U \longrightarrow V$ is a homeomorphism.*

Corollary. *If μ and ν are two norms on a finite dimensional vector space, then they give rise to the same topology.*

Proof. The identity function is a linear isomorphism, which must be continuous. This implies that every open set in the topology of ν is open in the topology of μ and vice-versa. ■

Given a finite dimensional normed vector space V, ν and a linear transformation $\alpha : V \longrightarrow V$, we define the *operator norm* of α with respect to ν by

$$\|\alpha\|_\nu = \sup\{\nu(\alpha(x)) : x \in V, \nu(x) = 1\}$$

This defines a norm on $\text{End}_\mathbb{K}(V)$, the set of all linear transformations $V \longrightarrow V$, which makes it a finite dimensional normed vector space over $\mathbb{K}$.

Proposition 1.2. *If μ and ν are two norms on a finite dimensional vector space V , then the topologies associated to $\text{End}_\mathbb{K}(V)$ with respect to μ and ν are the same.*

Inside $\text{End}_\mathbb{K}(V)$ we have the group of linear isomorphisms, $\text{GL}_\mathbb{K}(V) \subseteq \text{End}_\mathbb{K}(V)$, the *general linear group* of the vector space V .

Proposition 1.3. *Let U and V be two vector spaces of the same dimension. If $\varphi : U \longrightarrow V$ is a linear isomorphism, then it induces a continuous group isomorphism with a continuous inverse,*

$$\varphi_* : \text{GL}_\mathbb{K}(U) \longrightarrow \text{GL}_\mathbb{K}(V); \quad \varphi_*(\alpha) = \varphi \circ \alpha \circ \varphi^{-1}$$

Generally, given a finite basis for V , there is an associated linear isomorphism $V \cong \mathbb{K}^n$ for some n , a ring isomorphism $\text{End}_\mathbb{K}(V) \cong M_n(\mathbb{K})$ and a group isomorphism $\text{GL}_\mathbb{K}(V) \cong \text{GL}_n(\mathbb{K})$.

Theorem 1.4. *Let V be a finite dimensional vector space over $\mathbb{K}$ with dimension n and let $\varphi : V \longrightarrow \mathbb{K}^n$ be a linear isomorphism. Then $\varphi_* : \text{GL}_\mathbb{K}(V) \longrightarrow \text{GL}_n(\mathbb{K})$ is a continuous group isomorphism with a continuous inverse. Hence $\text{GL}_\mathbb{K}(V)$ is a matrix group.*

1.5 Matrix Exponential and Logarithm

We define the exponential and logarithm functions for matrices using their power series. Throughout this section, we assume $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

The *exponential function* is, $\exp : M_n(\mathbb{K}) \longrightarrow \text{GL}_n(\mathbb{K})$;

$$\exp(A) = \sum_{n \geq 0} \frac{1}{n!} A^n = I + A + \frac{1}{2!} A^2 + \frac{1}{3!} A^3 + \dots$$

Proposition 1.4. *Let $A \in M_n(\mathbb{K})$.*

- (i) *For $u, v \in \mathbb{C}$, $\exp((u + v)A) = \exp(uA) \exp(vA)$.*
- (ii) *$\exp(A)^{-1} = \exp(-A)$.*
- (iii) *If $B \in M_n(\mathbb{K})$ commutes with A , then $\exp(A + B) = \exp(A) \exp(B)$.*

We define the logarithm function by

$$\log : N_{M_n(\mathbb{K})}(I; 1) \longrightarrow M_n(\mathbb{K});$$

$$\log(A) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} (A - I)^n$$

Proposition 1.5. *The exp and log functions satisfy*

- (i) *if $\|A - I\| < 1$, then $\exp(\log(A)) = A$;*
- (ii) *if $\|\exp(B) - I\| < 1$, then $\log(\exp(B)) = B$.*

The functions exp and log are continuous and in fact infinitely differentiable on their domains. By continuity of exp at O , there is a $\delta_1 > 0$ such that

$$N_{M_n(\mathbb{K})}(O; r) \subseteq \exp^{-1} N_{GL_n(\mathbb{K})}(I; 1).$$

Since $\exp N_{M_n(\mathbb{K})}(O; r) \subseteq N_{M_n(\mathbb{K})}(I; e^r - 1)$, we can take $\delta_1 = \log 2$. Hence we have the following result.

Proposition 1.6. *The exponential function exp is injective when restricted to the open subset $N_{M_n(\mathbb{K})}(O; \log 2) \subseteq M_n(\mathbb{K})$. Hence it is locally a diffeomorphism at O with the local inverse log.*

1.6 Differential Equations in Matrices

Definition 1.4. *A differentiable curve in $M_n(\mathbb{K})$ is a function $\alpha : (a, b) \longrightarrow M_n(\mathbb{K})$ for which the derivative $\alpha'(t)$ exists for each $t \in (a, b)$. Here $\alpha'(t)$ is defined as an element of $M_n(\mathbb{K})$ by*

$$\alpha'(t) = \lim_{s \rightarrow t} \frac{1}{s - t} (\alpha(s) - \alpha(t)),$$

provided that the limit exists.

Consider the first order differential equation $\alpha'(t) = \alpha(t)A$, where α is assumed to be a differentiable curve.

Theorem 1.5. For $A, C \in M_n(\mathbb{R})$ with A non-zero, and $a < 0 < b$, the differential equation $\alpha'(t) = \alpha(t)A$, has a unique solution $\alpha : (a, b) \rightarrow M_n(\mathbb{R})$ for which $\alpha(0) = C$. Furthermore, if C is invertible, then so is $\alpha(t)$ for $t \in (a, b)$.

Proof. Consider the function $\alpha : (a, b) \rightarrow M_n(\mathbb{R})$; $\alpha(t) = \exp(tA)$. It is differentiable with

$$\alpha'(t) = \sum_{k \geq 1} k \frac{t^{k-1}}{(k-1)!} A^k = A \exp(tA)$$

Hence α satisfies the differential equation with boundary condition $\alpha(0) = I$. Clearly, $\alpha(t) = C \exp(tA)$ is one solution to $\alpha(0) = C$. If β is another such solution, then $\gamma(t) = \beta(t) \exp(-tA)$ satisfies

$$\begin{aligned} \gamma'(t) &= \beta'(t) \exp(-tA) + \beta(t) \exp(-tA)(-A) \\ &= \beta(t)A \exp(-tA) - \beta(t)A \exp(-tA) \\ &= 0. \end{aligned}$$

Hence γ is a constant function with $\gamma(t) = \gamma(0) = C$. Thus $\beta(t) = C \exp(tA) = \alpha(t)$. ■

1.7 One Parameter Subgroups in Matrix Groups

Chapter 2

Tangent Spaces and Lie Algebras

2.1 Lie Algebras

2.2 Curves and Tangent Spaces

2.3 Lie Algebras of Matrix Groups

2.4 $SO(3)$ and $SU(2)$

2.5 Complexification of Real Lie Algebra

Chapter 3

Lie Groups

3.1 Smooth Manifolds

3.2 Tangent Spaces and Derivatives

3.3 Lie Groups

3.4 Matrix Groups are Lie Groups

3.5 Not All Lie Groups are Matrix Groups

References

- [Baker, 2002] Baker, A. (2002). *Matrix Groups*. Springer Undergraduate Mathematics Series. Springer-Verlag.
- [Trapp, 2005] Trapp, K. (2005). *Matrix Groups for Undergraduates*. Student Mathematical Library. American Mathematical Society, second edition.